

Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -2(Adv-2025-P2-Model)** **Date: 13-05-2026**

Time: 02.30Pm to 05.30Pm **GTA-32** **Max. Marks: 180**

13-05-2026_Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT_Jee-Adv(2025-P2)_GTA-32_Syllabus,

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

MATHEMATICS : TOTAL SYLLABUS

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Name of the Student: _____ **H.T. NO:** | | | | | | | | |

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JEE-ADVANCED-2025-P2-Model

Time:3Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 4)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 5 – 8)	Questions with Multiple Correct Choice with partial mark	+4	-2	4	16
Sec – III(Q.N : 9 – 16)	NUMERICAL VALUE	+4	0	8	32
Total				**16**	**60**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 17 – 20)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 21 – 24)	Questions with Multiple Correct Choice with partial mark	+4	-2	4	16
Sec – III(Q.N : 25 – 32)	NUMERICAL VALUE	+4	0	8	32
Total				**16**	**60**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 33 – 36)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 37 – 40)	Questions with Multiple Correct Choice with partial mark	+4	-2	4	16
Sec – III(Q.N : 41 – 48)	NUMERICAL VALUE	+4	0	8	32
Total				**16**	**60**

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MATHEMATICS **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

1. Three circles each of radius 3 are drawn with centers at (0, 16), (3, 0) and (5, 8). A line of slope m passing through (3, 0) is such that the total area of the part of the three circles to one side of the line is equal to the total area of the three circles to the other side of it.

Then $\left|\frac{m}{8}\right|$ is equal to

A) 1 **B)** 2 **C)** 3 **D)** 4

2. If the coefficient of x^{78} in the expansion of $\left(1+x+2x^2+4x^4\right)^{20}$ is $\lambda\cdot 2^{40}$, then λ is equal to

A) 11 **B)** 10 **C)** 8 **D)** 4

3. A variable point P is on the circle $x^2+y^2=1$ on XY–plane. From point P perpendicular PN is drawn to the line $L: x-2=y-2=-z$, then minimum length PN drawn is

A) $\sqrt{\frac{9}{2}-2\sqrt{2}}$ **B)** $\sqrt{3-4\sqrt{2}}$ **C)** $\frac{2\sqrt{2}-1}{\sqrt{3}}$ **D)** $\frac{2\sqrt{2}-1}{3}$

4. If $x,y,z,w\in R$ satisfy the following system of equations

$x+y+z+w=1; x+2y+4z+8w=16;\ x+3y+9z+27w=81$ and

$x+4y+16z+64w=256$, then the pair which has H.C.F as 2 is

A) $(|w|,|z|)$ **B)** $(|z|,|y|)$ **C)** $(|y|,|x|)$ **D)** $(|z|,|x|)$

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SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
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Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.
For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

5. Which of the following is/are **TRUE**?

A) Let $\alpha = 5\beta$ and $\sin(\alpha+\beta)+\sin 2\beta = 3\cos 4\beta$ then the value of $\frac{\sin(\alpha-3\beta)}{1-\tan^2 2\beta} = \frac{3}{4}$ (where $\sin 2\beta > 0$).

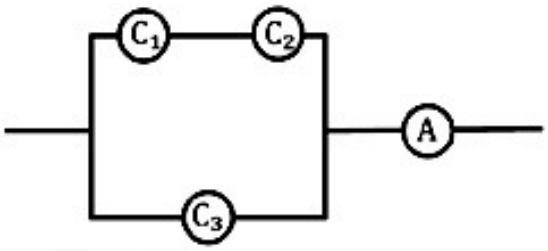
B) If $\alpha = 5\beta$ and $2\operatorname{cosec}\frac{\alpha-\beta}{4} = \frac{1}{\cos\alpha}+\frac{1}{\sin\beta}$, then the number of values of β in $(0,\pi)$ is 5.

C) Let $\alpha = 5\beta$ and $\sin(\alpha+\beta)+\sin 2\beta = 3\cos 4\beta$ then the value of $\frac{\sin(\alpha-3\beta)}{1-\tan^2 2\beta} = \frac{4}{5}$ (where $\sin 2\beta > 0$).

D) If $\alpha = 5\beta$ and $2\operatorname{cosec}\frac{\alpha-\beta}{4} = \frac{1}{\cos\alpha}+\frac{1}{\sin\beta}$, then number of values of β in $(0,\pi)$ is 4.

6. An electrical apparatus A with three components C_1, C_2 and C_3 as in the diagram. The components works independently of each other with probabilities $\frac{8}{10}, \frac{9}{10}$ and $\frac{3}{4}$. For apparatus to be operational current flows from left to right. Let X denote the event that apparatus is operational and X_1, X_2, X_3 denotes respectively the events that the components C_1, C_2 and C_3 are functioning. Which of the following is (are) **TRUE**?

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A) P (exactly two components are functioning / X) = 1.

B) If P(X) is $\frac{a}{b}$ (a, b are co–prime) then b – a is 7.

C) $P(X/X_3)=1$

D) $P(X/X_2)=19/20$

7. Which of the following is/are **TRUE**?

A) In any ΔABC, minimum value of $\sum\frac{\sqrt{\sin A}}{\sqrt{\sin B}+\sqrt{\sin C}-\sqrt{\sin A}}$ is 3.

B) If $\sec A\tan B+\tan A\sec B=30$ then $\sec A\sec B+\tan A\tan B=\sqrt{901}$.

C) If $\sec A\tan B+\tan A\sec B=30$ then $\sec A\sec B+\tan A\tan B=\sqrt{899}$.

D) Three lines $y-z-1=0=x,\ z+x+1=0=y,\ x-z-1=0=y$ intersect the xy plane at U,M,R then orthocenter of triangle UMR is $(0,1,0)$.

8. Let the function $f:(-\infty,\infty)\to\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$ be given by $f(x)=\sin^{-1}\left(\ell og_3\left(\frac{x^2-x+1}{x^2+x+1}\right)\right)$, then

A) $f\left(\frac{1}{x}\right)=-f(-x)$

B) $f(x)$ is a strictly increasing function in $(-\infty,\infty)$

C) $f(x)$ is a surjective function

D) $f(x)$ is a injective function

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SECTION-III
(NUMERICAL VALUE)

- This section contains **EIGHT (08)** questions stems.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
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- **Zero Marks:0** in all other cases

9. On an ellipse $\frac{x^2}{64}+\frac{y^2}{9}=1$, tangents drawn at $P_1,P_2,P_3,.....,P_n$ intersects the major axis at $T_1,T_2,T_3,.....,T_n$ respectively. If the value of $\sum_{i=1}^{n}\frac{Area(\Delta P_iT_iS)\cdot Area(\Delta P_iT_iS')}{(P_iT_i)^2}=18$, then n is equal to _____

10. Let $f(x)$ be a function defined on R such that $f'(x)=f'(3-x)\forall x\in[0,3]$ with $f(0)=-32$ and $f(3)=46$, then find the value of $\frac{1}{3}\int_0^3 f(x)dx$

11. Consider A and B as 2×2 matrices with determinant equal to 1, then $\left|tr(AB)-tr(A)\cdot tr(B)+tr\left(AB^{-1}\right)\right|+2$ is _____

12. If a,b,c are three positive real numbers then the minimum value of $\frac{a+3c}{a+2b+c}+\frac{4b}{a+b+2c}-\frac{8c}{a+b+3c}$ is $\alpha+\beta\sqrt{2}$ (where $\alpha,\beta\in Z$), then $|\alpha+\beta|$ is equal to _____

13. Let $AB: y=x$ be a chord of the parabola $y^2=-4x+4$ and P be any variable point on the ellipse $x^2+9(y-4)^2-9=0$. Given that Δ_1 and Δ_2 be the minimum and maximum area of triangle ABP respectively. If $\lambda=\Delta_1\Delta_2$, then $\frac{\lambda}{4}$ equals

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14. $\left|\lim\limits_{x\to\frac{\pi}{2}^-}(1+\tan x)\left\{(1+\tan x)\ell n\left(\frac{1+\tan x}{2+\tan x}\right)+1\right\}\right|$ is equal to

15. If $f(x)$ is defined on $[-3,3]$ by $f(x)=4x^3-2x+1$ and $g(x)=\frac{f(x)-f(-x)}{|x|+2}$, then value of $\int\limits_{-3}^{3} g(x)dx$ is equal to

16. Let $f(x)=\left(1+x+x^2+.....+x^n\right)^2$. If $C(r,n)$ denotes coefficient of x^r in the polynomial expansion of $f'(x)$. Then the value of $2026\left(\sum\limits_{r=0}^{2024}\frac{1}{C(r,2026)}\right)$ is

{Note: $f'(x)$ denotes derivate of $f(x)$}

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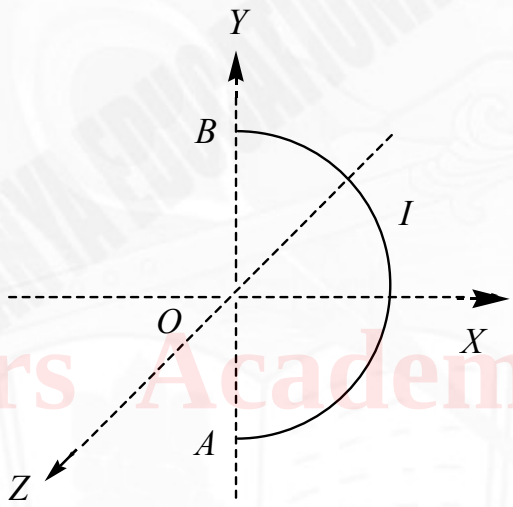
PHYSICS **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
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17. A conductor carrying current I is in the form of a semicircle AB of radius R and lying in xy-plane with it's centre at origin as shown in the figure. Which of the following is the magnitude of $\int \bar{B}.\bar{dl}$ for the circle $3x^2+3z^2=R^2$ in xz-plane due to current in curve AB?

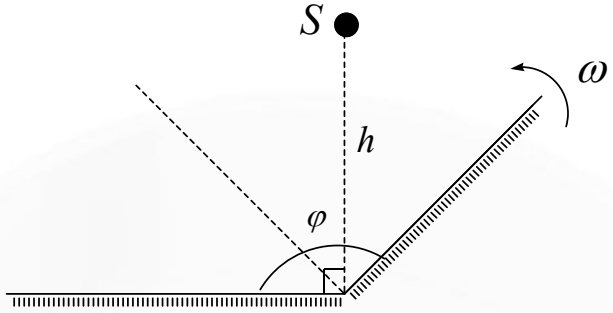


A) $\frac{\sqrt{3}}{2}\mu_0 I$ **B)** $\left(\sqrt{3}+1\right)\frac{\mu_0 I}{2}$ **C)** $\left(2-\sqrt{3}\right)\frac{\mu_0 I}{2}$ **D)** Zero

18. The angle between two flat mirrors is changed by rotating one of the mirrors around the edge of another with a constant angular velocity $\omega = 1.5$ deg/ sec. Source of light S is placed as shown in the figure at a distance h=10cm. At the initial moment mirrors were in the same plane $\left(\varphi = 180^\circ\right)$. After what minimum time (in seconds), 3 images will be formed by the mirrors.

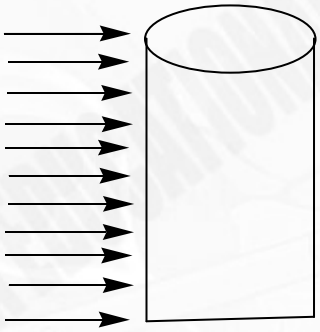
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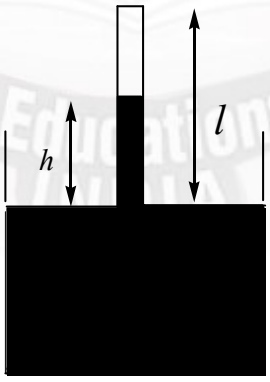
A) 20 **B)** 30 **C)** 40 **D)** 60

19. Find the force exerted by a light beam of intensity I, incident on a cylinder (height h and base radius r) placed on a smooth surface as shown in figure if surface of cylinder is having reflection coefficient 0.8 (assume no transmission)



A) $\frac{38}{15}\frac{Irh}{c}$ **B)** $\frac{23}{15}\frac{Irh}{c}$ **C)** $\frac{70}{15}\frac{Irh}{c}$ **D)** $\frac{58}{15}\frac{Irh}{c}$

20. When a vertical capillary of length 'l' with the sealed upper end was brought in contact with the surface of liquid, the level of this liquid rose to the height 'h'. The liquid density is 'ρ', the inside diameter of the capillary is 'd' , the contact angle is θ, the atmospheric pressure is P_0. The surface tension of the liquid is (Temperature in this process remains constant.)



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A) $\frac{d}{4\cos\theta}\left[\frac{P_0 h}{l-h}+\rho g h\right]$ **B)** $\frac{d}{2\cos\theta}\left[\frac{P_0 h}{l-h}+\rho g h\right]$

C) $\frac{d}{4\cos\theta}\left[\frac{P_0 l}{l-h}+\rho g h\right]$ **D)** $\frac{d}{2\cos\theta}\left[\frac{P_0 l}{l-h}+\rho g h\right]$

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
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choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

21. In a long wire of square cross-section of side length 'l', current density varies with distance 'x' from one edge of cross-section as $J = ae^{bx} A/m^2$ where a and b are positive constants, then the correct option(s) is/are

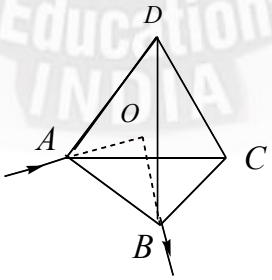
A) Current flowing in the wire is $i = \frac{al}{b}\left(e^{bl}-1\right)$

B) Current flowing in the wire is $i = \frac{al}{b}\left[e^{-bl}-1\right]$

C) Current flowing in the wire is not uniform along the length

D) Current flowing in the wire is uniform along the length

22. A regular tetrahedron frame is made up of homogenous resistance wire of uniform cross section. Current I enters vertex A through a long, straight wire directed towards the centre O of the tetrahedron and is conducted away through vertex B in the same way, as illustrated in the figure. Choose **CORRECT** statement(s)



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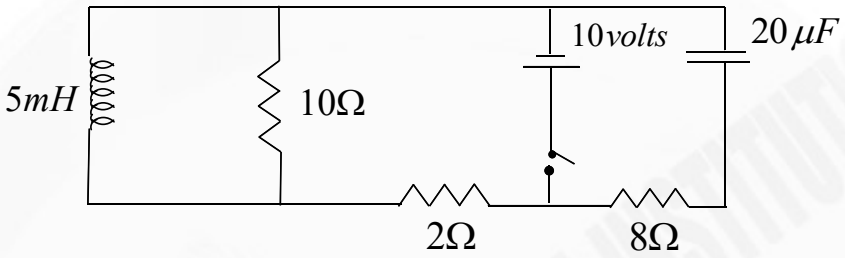
A) There is no current through wire DC because point D and C are equipotential

B) Current through wire AB is I/2

C) Current through wire AC is I/4

D) Net magnetic field at point O is zero

23. In the given circuit at t=0, the switch is closed.



A) The current through the $10\,\Omega$ resistor at any instant of time $t(0<t<\infty)$ will be

$\frac{5}{6}\times e^{-(1000/3)t}\,A$

B) The current through the $10\,\Omega$ resistor at any instant of time $t(0<t<\infty)$ will be

$\frac{1}{6}\times e^{-(1000/3)t}\,A$

C) The energy stored in capacitor as $t\to\infty$ will be 1 millijoule

D) The energy store in inductor as $t\to\infty$ will be 62.5 millijoules

24. A transverse harmonic wave is propagating along a taut string. Tension in the string is 50 N and its linear mass density is 0.02 kg/m. The string is driven by a 80 Hz oscillator tied to one end, oscillating with amplitude of 1 mm. The other end of string is terminated so that all the wave energy is absorbed and there is no reflection. From this given data, choose the correct option (s)

A) Average power of the oscillator is 0.126 W

B) If tension in the string is quadrupled and power of oscillator remains same, then new amplitude of oscillation of string is $\frac{1}{\sqrt{2}}mm$

C) Average energy of the wave on a 1 m long segment of the string is approximately 2.5 mJ

D) Average power of the oscillator is 0.252 W

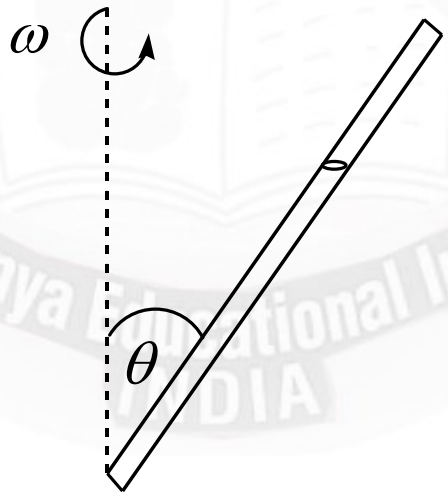
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SECTION-III
(NUMERICAL VALUE)

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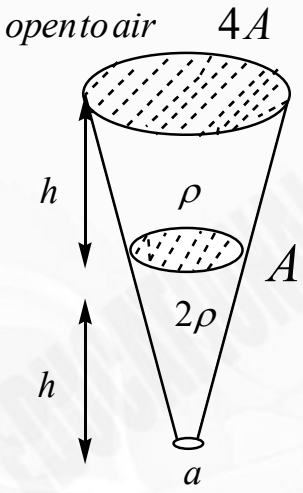
25. A neutron strikes a hydrogen atom which is initially at rest and free to move. The initial speed of the neutron is v_0. After the collision both neutron and the H atom move making 30° angle with the initial velocity of the neutron. Also the H atom is found to get exited to first excited state. If the initial KE of neutron is 'x' times first excitation energy of the H atom, find the value of 'x' (assume mass of neutron=mass of H atom)

26. A closed tube of length 'l' completely filled with water has a small air bubble trapped in it. When the tube is held at an angle '$\theta = 30^\circ$'with the vertical and rotated at a constant angular velocity 'ω' about the vertical axis through its lower end, the bubble settles at some intermediate position in the tube. The distance of the bubble from the lower end of the tube is $\frac{a\sqrt{bg}}{\omega^2}$. (Acceleration due to gravity is 'g') Find (a+b)

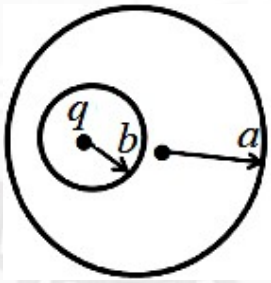


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27. Two immiscible liquids are filled in conical flask as shown in figure. The areas of cross sections are shown, a hole of area 'a' is made in lower end of cone. The speed of liquid flow from hole is $\sqrt{\frac{(n\times gh)}{1-\frac{17a^2}{32A^2}}}$. Then n is $(a << A)$



28. A point charge q is placed at the centre of a spherical cavity of radius 'b' inside a neutral conducting sphere of radius 'a'. Then total electrostatic potential energy is $\left(\frac{kq^2}{xa}-\frac{kq^2}{ya}\right)$ then $(x\times y)$ is. (ignore self energy of point charge)

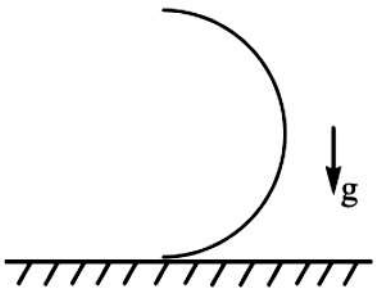


29. A cube made of homogeneous materials is kept on a horizontal surface. If the ratio of strain energy stored in lower one third to the upper one third of the cube is Z, then what is the value of Z?

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30. A uniform disc of mass m and radius R is on a rough horizontal surface (with circular surface in contact). Coefficient of static friction between disc and the rough surface is μ. The minimum amount of torque required to cause rotation of the disc about an axis passing through its centre and perpendicular to its plane is $\frac{n}{3}\mu mgR$. The value of n is

31. A uniform thin hollow hemispherical shell of mass m and radius R is released from rest with its diametrical plane vertical on a very rough horizontal surface. It doesn't slip on the floor during its subsequent motion. At an instant when the shell achieves its maximum value of kinetic energy, the value of normal reaction on it from the floor is xmg, where g is acceleration due to gravity, then find the value of x?



32. If you hang a long uniform inextensible rope, the longest hanging length is $l_0 = 2m$ that does not break due to its own weight. Now you cut a length of the rope and place it straight on a horizontal frictionless table with a little length hanging over the edge so that it begins to slide down when released. What maximum length (in meters) can this piece have so that it does not break during sliding?

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CHEMISTRY **Max Marks: 60**

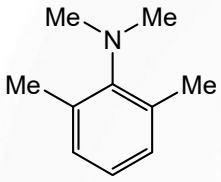
SECTION–I
(SINGLE CORRECT ANSWER TYPE)

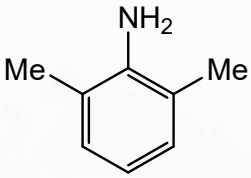
This section contains **FOUR (04)** questions.

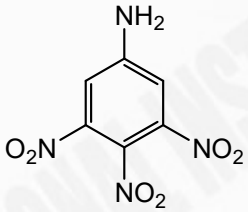
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

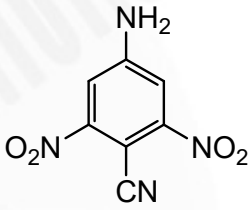
Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

33. The correct order of basic strength of following amines is:

I) 

II) 

III) 

IV) 

A) I > II > III > IV **B)** I < II < III < IV

C) II < I < IV < III **D)** I > II > IV > III

34. A real gas follows the Van der Waals equation: $\left(P + a/V_m^2\right)\left(V_m - b\right) = RT$.

At a specific high-temperature regime, the molar volume V_m is large enough that the term a/V_m^2 becomes negligible, but V_m is not large enough to ignore b. If the collision frequency (Z) of this gas is compared to an ideal gas $\left(Z_{ideal}\right)$ at the same P and T, and we assume the mean molecular speed remains unchanged, which statement correctly describes the system?

A) The compressibility factor $Z_c < 1$; the actual volume available for molecular motion is $V_m + b$.

B) The compressibility factor $Z_c = 1 + \left(Pb/RT\right)$; the collision frequency Z is higher than Z_{ideal} because the excluded volume increases effective density.

C) The pressure correction term a/V_m^2 represents internal pressure which increases wall collision frequency.

D) The ratio $Z/Z_{ideal} = \left(1 - b/V_m\right)$; collisions decrease because molecules occupy more space.

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35. Identify the incorrect statement regarding polymers from the options given below:

A) Ethene polymerized at 350–570 K and 1000–2000 atm in the presence of a peroxide initiator yields high-density polyethylene.

B) Teflon is obtained by polymerizing tetrafluoroethene in the presence of a persulfate catalyst under high pressure.

C) Polyvinyl chloride (PVC) is a thermoplastic polymer.

D) Polymerization of chloroprene produces neoprene as the polymer.

36. An ore containing galena (PbS) and sphalerite (ZnS) is concentrated by froth flotation. Sodium cyanide is added as a depressant, leading to the formation of a soluble coordination complex (X) in the aqueous phase, while the other sulphide floats. A sample of the aqueous solution containing (X) is subjected to qualitative analysis:

Test 1: Passing H_2S gas directly into the solution produces no precipitate.

Test 2: The solution is first acidified with dilute $HC\ell$, boiled to expel dissolved gases, and then treated with H_2S. A white precipitate (Y) is formed.

Which pair correctly identifies (X) and (Y)?

A) $(X)=\left[Pb(CN)_4\right]^{2-};(Y)=PbS$

B) $(X)=\left[Zn(CN)_4\right]^{2-};(Y)=ZnS$

C) $(X)=\left[Pb(CN)_2\right];(Y)=PbC\ell_2$

D) $(X)=\left[Zn(CN)_4\right]^{2-};(Y)=S(colloidal)$

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SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.

For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

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37. In the following reactions P, Q, R and S are the major products

$$CH_3CH_2CH(CH_3)CH_2CN \xrightarrow[ii)CH_3MgBr, then\ H_2O]{i)CH_3MgBr, then\ H_3O^+} P$$

$$C_6H_6 + CH_3COCl \xrightarrow[ii)\ PhMgBr,\ then\ H_2O]{i)\ Anhy.AlCl_3} Q$$

$$CH_3CH_2COCl \xrightarrow[ii)CH_3MgBr,\ then H_2O]{i)\frac{1}{2}(PhCH_2)_2Cd} R$$

$$PhCH_2CHO \xrightarrow[iii)HCN \\ iv)H_2SO_4,\Delta]{i)CH_3MgBr, then\ H_2O \\ ii)CrO_3, dil.H_2SO_4} S$$

The correct statement(s) about P, Q, R and S is (are)

A) Both P and Q have chiral carbon(s)

B) Both Q and R have asymmetric carbon(s)

C) P has asymmetric carbon(s), S does not have any asymmetric carbon

D) Both P and R have chiral carbon(s)

38. On addition of a neutral, monodentate organic ligand L to $\left[M\left(H_2O\right)_6\right]^{2+}_{(aq)}$ the following equilibrium is established:

$$\left[M\left(H_2O\right)_6\right]^{2+}_{(aq)} + 6L(\ell) \rightleftharpoons \left[M\left(L\right)_6\right]^{2+}_{(aq)} + 6H_2O(\ell)$$

The following observations are made:

1) The equilibrium constant K decreases with increase in temperature.

2) $\left[M\left(L\right)_6\right]^{2+}$ is diamagnetic, whereas $\left[M\left(H_2O\right)_6\right]^{2+}$ is paramagnetic with a spin-only magnetic moment of 4.90 BM.

3) The reaction is spontaneous at 298 K under standard conditions.

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4) ΔG° value becomes less negative with increase in temperature and ΔH° and ΔS° are constants.

5) Ligand L is a liquid at 298 K, immiscible with water, and its standard state is the pure liquid.

Which of the following statements are correct?

A) The reaction is enthalpy-driven and ΔS° is negative.

B) The metal ion can be Fe^{2+} and ligand L produces a low-spin d^6 configuration.

C) ΔH° is negative mainly due to the greater crystal field stabilization energy of $\left[M(L)_6\right]^{2+}$ compared to $\left[M(H_2O)_6\right]^{2+}$.

D) Changing the standard state of L from pure liquid to 1 M aqueous solution changes the magnitude of ΔS°.

39. Select the correct chemical changes:

A) $Cu + dil.H_2SO_4 + \frac{1}{2}O_2(g) \longrightarrow CuSO_4 + H_2O$

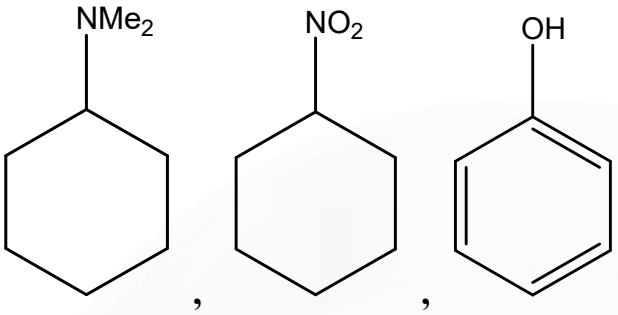
B) $Cu + dil.HNO_3 \longrightarrow Cu(NO_3)_2 + N_2O(g)$

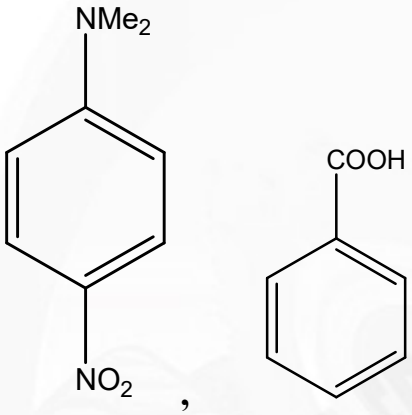
C) $CuSO_4(aq) + KCN(excess) \longrightarrow K_2\left[Cu(CN)_4\right]$

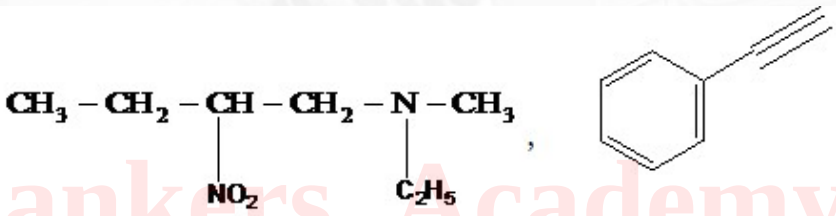
D) $CuSO_4(aq) + NH_4OH \longrightarrow Cu(OH)_2(S) + (NH_4)_2\,SO_4$

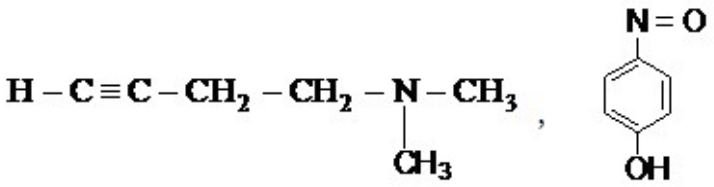
40. When a mixture containing set of organic compounds was divided equally in three parts and each part was treated with three different reagents separately only one organic compound responds to one reagent at a time. If the reagents chosen are $PhSO_2C\ell$ followed by KOH (ii) $Zn / NH_4C\ell$ followed by Tollen's reagent (iii) Na metal. Find which options given must have more than one compound of original mixture.

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A) 

B) 

C) 

D) 

SECTION-III
(NUMERICAL VALUE)

- This section contains **EIGHT (08)** questions stems.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated <u>according to the following marking scheme:</u>
- ***Full Marks: +4*** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

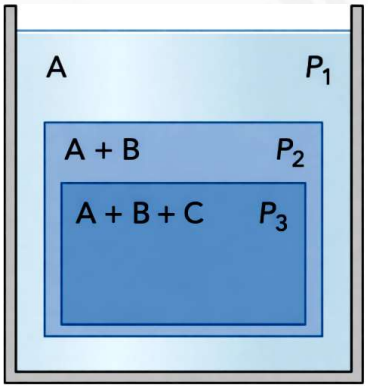
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41. For each row in the table below, select the option number (1–4) that correctly satisfies the stated property. Write the selected option numbers in order (row 1 → row 3) to form a three-digit number. Report this number as your final answer.

No of the row	Property	1	2	3	4
1	Highest positive charge per molecule at pH = 6	Lysine	Alanine	Glutamic acid	Serine
2	Most stable hydride	SiH_4	GeH_4	SnH_4	PbH_4
3	Smallest K_{sp}	NiS	CuS	ZnS	Al_2S_3

42. Three nested liquid compartments are separated by selective membranes and attains osmatic equilibrium.



Cell 1 contains pure solvent A at pressure P_1.

Cell 2 contains a dilute solution of A + B at pressure P_2.

Cell 3 contains a dilute solution of A + B + C at pressure P_3.

The membrane between cell 1 and cell 2 allows only solvent A to pass but blocks solute B.

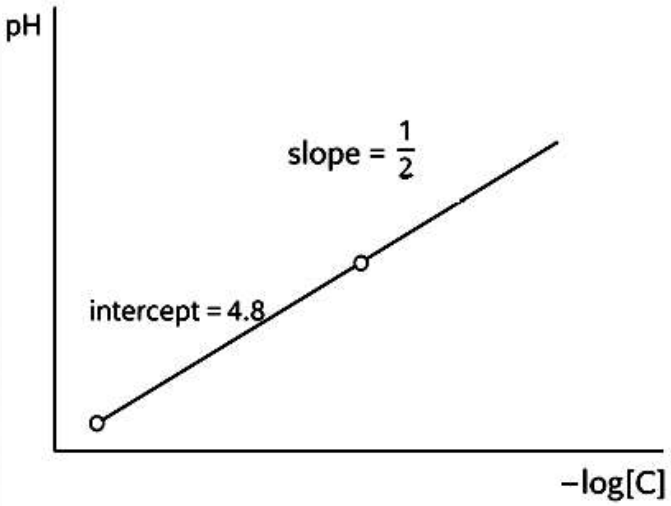
The membrane between cell 2 and cell 3 allows A and B to pass but blocks solute C.

Assume ideal dilute behavior $(X_A >> X_B, X_C)$ where X represents mole fraction.

If $[B] = 2\,mol\,L^{-1}[C] = 3\,mol\,L^{-1}.RT = 10\,L.bar\,mol^{-1}$, find the pressure difference, $(P_3 - P_1)$ in bar.

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43. A salt XY, formed from a strong acid and a weak base (XOH), is dissolved in water at 298 K. For small degree of hydrolysis (h << 1), the qualitative graph of pH versus −log C is shown below.



If 80.0 g of XY is used to prepare 2.0 L of solution, and XY crystallizes in a CsCl-type lattice with ionic radii (in pm) $r_+ = 120\sqrt{3}, r_- = 80\sqrt{3}$, calculate the value of:

M/(10 ρ) − pH at C = 0.1 M

where M is molar mass (g/mol) and ρ is density $\left(g/cm^3\right)$. Use $N_A = 6\times10^{23}$

44. Consider the following complexes:

(i) $\left[Cr(EDTA)\right]^-$ (ii) $\left[Ru(en)_3\right]^{2+}$ (iii) $\left[Pt(gyl)C\ell_2\right]^-$

(iv) $cis-\left[Cr(C\ell)_2(ox)_2\right]^{3-}$ (v) $trans-\left[Pt(en)(NH_3)_2(py)_2\right]^{4+}$

Find the value of the expression $|p\times q|^2$.

Where,

p = total number of optically active complexes.

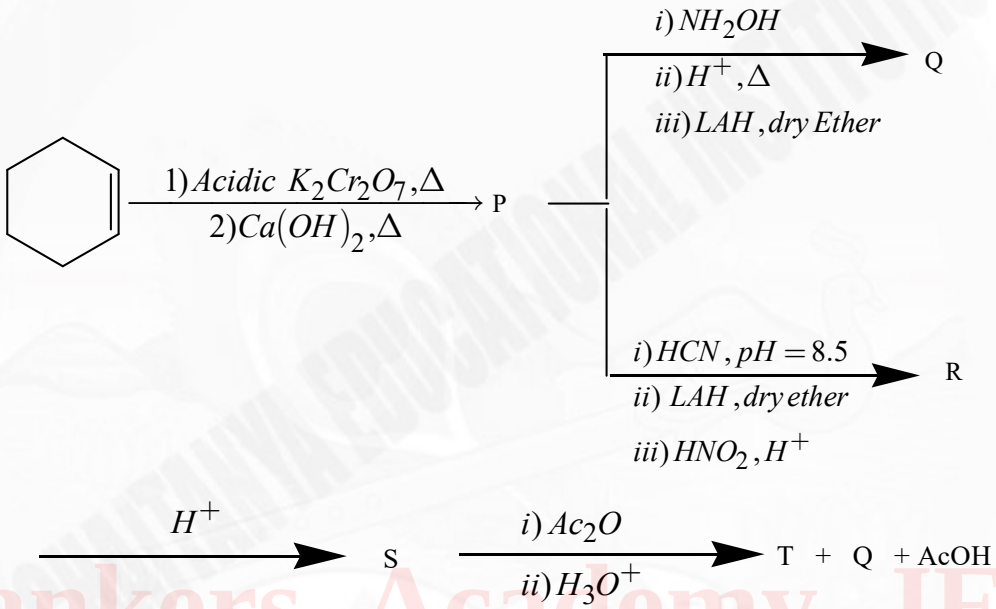
q = total number of optically inactive complexes.

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45. The electrode potential of $2Ag(s)+S^{2-}(aq)\rightarrow Ag_2S(s)+2e^-$ in a solution buffered at pH = 3 and which is also saturated with $0.1M-H_2S$, is 142 mV. What is the value of $E^o_{Ag^+|Ag}$ in Volts ?

(Given: for $H_2S, K_{a_1}=10^{-8}, K_{a_2}=10^{-13}, K_{sp}$ of $Ag_2S=4\times10^{-48}$, log 2 = 0.3, 2.303 RT/F = 0.06)

46. Consider the following scheme of reaction.



$$R + Q \xrightarrow{H^+} S \xrightarrow[ii)\,H_3O^+]{i)\,Ac_2O} T + Q + AcOH$$

Given atomic mss data: H = 1, C = 12, N = 14, O = 16.

What is the molecular mass of T in g/mol?

47. When a stationary hydrogen-like ion in the excited state n = 2 makes a transition to the ground state n = 1, the momentum of the emitted photon is P_{ph}. If the same ion undergoes a transition from n = 3 to n = 1, the emitted photon has momentum P'_{ph}. If $P'_{ph}/P_{ph}=X/27$, find the integer value of X.

48. Consider all the esters of molecular formula $C_7H_{14}O_2$, excluding stereoisomers. They are all treated with an excess of methyl magnesium iodide and then acidified to give alcohols. Each ester produces two different alcohols [excluding stereoisomers], both of which react with $Br_2/NaOH$ to give bromoform. The number of esters, satisfying all these criteria, is _____.

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63 RANK SAMYAJYOTI BISWAS HT. No. 255058456*	64 RANK AARUSH ANAND HT. No. 251006176*	72 RANK RUSHMITH BANDARI HT. No. 256168048	78 RANK KORIKANA RASAGNYA HT. No. 256057046	87 RANK LAKSHYA SHARMA HT. No. 252070079*	91 RANK AVANEESH BANSAL HT. No. 251113130*	95 RANK KAVYA AGGARWAL HT. No. 252079121*

BELOW 100 ALL INDIA OPEN CATEGORY RANKS	29	BELOW 500 ALL INDIA OPEN CATEGORY RANKS	113	BELOW 1000 ALL INDIA OPEN CATEGORY RANKS	205	BELOW 1000 ALL INDIA CATEGORY RANKS COUNT	745	NUMBER OF QUALIFIED RANKS	4,212

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